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Peer Review File

Medical Multimodal Multitask Foundation Model for Lung Cancer Screening

Corresponding Author: Professor Ge Wang

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
Strength:
• This is the first multimodal multitask pretrained foundational for the LDCT lung cancer screening.
• The authors demonstrate superior performance for nearly all included tasks over previous SOTA, including some core tasks like lung cancer risk, incidental lung abnormalities, and CVD risk predictions.

Weakness:
• The definition of CVD mortality is oversimplified, and many details are missing. Take NLST for example, the mortality follow-up spans over a median of 6.5 years for the initial close date of December 31, 2009, and latterly extended to a median of 12.3 years to the end of 2015 (PMCID: PMC6764895). So, the first question is which end-date/follow-up periods the authors considered to define the CVD mortality. Moreover, the mortality follow-up data is a typical right censored time-to-event data with significant loss of follow-up. Thus, it is an over-simplification to take CVD mortality as a binary label. As lung cancer risk prediction tasks were stratified from 1 to 6 cut-off years, the authors should explain why not conducting similar cut-off time stratified analysis for CVD diagnosis and mortality risk predictions.
• The pre-trained imaging encoder can only take imaging patches of lungs and hearts as input. This should be considered as a limitation, as this will pose restrictions on other potential opportunistic assessment of body sections depicted in lung screening LDCT such as liver fat, bone mineral density, and body compositions (10.1148/radiol.222044, 10.1002/jbmr.4187, 10.1148/radiol.222937, 10.1016/j.clinimag.2019.12.009)
• To encode the clinical meta data such as demographic, lifestyle, and disease diagnostic history, the authors created a clinical text following certain predefined rules and used this clinical text as condition to obtain the text embeddings forwarded to the task encoder. Since these clinical texts followed predefined rules, this fundamentally differed from the unstructured text in natural language and pose the question that whether or not this step is necessary comparing to simply encode these clinical metadata as an array.
• Fig.1 can be improved to clearly indicate the input / output for the training and inference/test steps. The dashed vertical line makes it confuse as to what types of data were used as training and inference.
• Line 384-385 – “We used the method developed in WFUSM as the reference”: missing a reference.
• Line 712-714: in addition to the number of V100 GPUs used, could you also provide the computing time?

(Remarks on code availability)
Detailed instructions are provided in the README file. There is demo case provided. I see no difficulties to reproduce the results following the instructions.

Reviewer #2

(Remarks to the Author)
The manuscript describes the development of a lung cancer screening multimodal foundation model that learns, in a multitask setting, how to extract information from both CT scans and clinical reports in the form of text. First of all, I believe that the application of innovative deep learning breakthroughs in real-world and complex domains such as the clinical environment is extremely valuable, not only to assess how current models behave under real-world scenarios but also to more easily expose certain failure modes that are easily covered by typical large-scale training.

Considering the field's state-of-the-art, the authors' analysis focuses on the fact that most of the literature is limited to single-modality and single-task (SM-ST) approaches, which I agree. However, I think that a deeper elaboration on why including different data modalities and training for several tasks at once brings any benefits when compared with SM-ST counterparts is still missing (more direct concerns related to this in the points below).

Overall, I found the methodology interesting, well-reported and with enough detail to ensure reproducibility. However, there are some concerns I believe the authors should elaborate more on:
- Is it possible that, for certain questions, irrelevant input information is given for the model to provide a prediction? For instance, Table 1 shows that for questions like "Is there any lung nodule" or "Where is the lung nodule", several different sources of text inputs are given alongside the CT exam. In cases like this, isn't it enough to look at imaging information? The authors mention this subject in l.431, but it remains unclear whether you agree or not that different modalities should be provided even if they're not relevant a priori.
- Are there any implications that result from treating clinical information that possibly comes from tabular data (hence being explicitly symbolic and represented in a non-ambiguous way) as free text? For instance, can your specific sentence format result in some sort of overfitting in the way that different sentences with equivalent meaning would cause the model to fail?
- The reported results demonstrate that, overall, AUC values increase for MT-MM variants. However, I think the authors could give a more critical perspective when balancing the advantages in terms of performance values - which in some cases are small - with the extra effort of processing multiple datasets and creating adequate tasks.
- The heatmaps in Figure 6 show that the presence of different text tokens is "important" for the model's decision, even though some of them are apparently useless by themselves (e.g. "brother", "142"). I think the authors should analyze this issue when describing the figure.
- Even though the analogy to human learning processes can be easily raised, can it be explicitly shown that information from multiple sources is actually being composed and learned in a complementary way? Is there any constraining effect on the kind of representations that are captured from image-based models when encoding multiple sources of information in parallel instead of only using images? When performing multiple tasks, is it possible to demonstrate that pieces of knowledge are being reused? If so, which ones? I think questions like these should be deeply explored in approaches like M3FM.
- Finally, the ablation studies performed are valuable, and the comparison with different models also highlights situations where performance gaps are larger and smaller, which is useful.

In general, even though the authors conducted several experiments, I think that showing a deeper analysis of the behavior of an MT-MM approach besides quantitative performance values is necessary to conclude that these training settings bring the benefits that are raised when motivating.

(Remarks on code availability)

Reviewer #3

(Remarks to the Author)
This manuscript details the development of multimodal multitask foundation model in the setting of lung cancer screening with low dose chest CT. The model is able to perceive multimodal data from CT and other clinical data sources and perform various tasks within the LCS workflow--including lung nodule detection and characterization, lung cancer risk prediction, cardiovascular disease diagnosis and mortality prediction, and Lung-RADS reporting/categorization, among others mimicking the multiple tasks a radiologist reading a LDCT for LCS screening would typical perform.

The model was developed from robust data sources (NLST and MIDRC) and tested on data from two large medical centers. The authors report that the model outperforms previous state-of-the-art models, including the Sybil models.

The manuscript is overall well-written. Methodology is sound and results and discussion are robust and appropriate. The model has promise in leveraging AI in optimizing lung cancer screening implementation and other multimodal/multitask programs in the future.

(Remarks on code availability)

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
My comments have been addressed comprehensively in the revision. I have no further comments to the authors.

(Remarks on code availability)

Reviewer #2

(Remarks to the Author)
I appreciate the authors' effort to analyze and answer my comments.
All the concerns previously raised were thoroughly clarified, hence I have no further comments.

(Remarks on code availability)

Point-by-Point Responses to Reviewer #1's Comments

Please note that all figure, table, and reference numbers mentioned correspond to those in the revised manuscript.

Strength:

• This is the first multimodal multitask pretrained foundational for the LDCT lung cancer screening.
• The authors demonstrate superior performance for nearly all included tasks over previous SOTA, including some core tasks like lung cancer risk, incidental lung abnormalities, and CVD risk predictions.

Response 1.0: We appreciate the reviewer's excellent high-level summarization of our work.

Weakness:

1. The definition of CVD mortality is oversimplified, and many details are missing. Take NLST for example, the mortality follow-up spans over a median of 6.5 years for the initial close date of December 31, 2009, and latterly extended to a median of 12.3 years to the end of 2015 (PMCID: PMC6764895). So, the first question is which end-date/follow-up periods the authors considered to define the CVD mortality. Moreover, the mortality follow-up data is a typical right censored time-to-event data with significant loss of follow-up. Thus, it is an over-simplification to take CVD mortality as a binary label. As lung cancer risk prediction tasks were stratified from 1 to 6 cut-off years, the authors should explain why not conducting similar cut-off time stratified analysis for CVD diagnosis and mortality risk predictions.

Response 1.1: Thanks for the very constructive comment. For a fair comparison with the existing CVD mortality risk prediction models, we had used the same dataset defined in Ref. [16]. In our revision, we have further checked the dataset and added the key information that "*… the intervals between screening CT and CVD mortality are in the range from 11 days to 2,619 days (within 8 years)...*" Inspired by the reviewer's comment, "*For the CVD mortality risk prediction task, we further stratified the binary risk into 1-6 cut-off year risks following [14].*" As for lung cancer risk prediction, we have defined CVD mortality risk labels for each of 6 years, taking into consideration the screening date, last follow-up date, death date, and cause of death. With the stratified labels, we have further trained and evaluated our M3FM and the SOTA models on the new dataset. The methods for labeling binary outcomes and 1–6 cut-off years are summarized in **Supplementary Table 2**, with the relevant part of the table listed below. **Supplementary Table 6** contains AUC (%) and 95% CI results of different models for stratified CVD mortality risk prediction, with part of it listed below. "*On average, M3FM enhanced the 1-6 year CVD mortality risk prediction performance by 14.22% in AUC compared to the previous best model (see Supplementary Table 6).*" and "*Similarly, for multi-year CVD mortality risk prediction, the multimodal model outperformed the single-modality model by 5% on average as shown in Supplementary Table 6.*" The distributions of CVD mortality risk labels are summarized in **Supplementary Table 7**, again part of it copied below for your convenience.

***Part of* Supplementary Table 2** *Labeling methods for different tasks. The ICD-10 codes for CVD-related causes of death include I10, I11, I12, I13, I20, I21, I24, I25, I63, I64, I65, I66, I67, I70, I71, I72, and I73.*

Tasks	**Labeling Methods**
CVD mortality risk within n years N ~ [1, 6]	1. The lung cancer risk is 0 if no pathology confirmed lung cancer within n years and the follow-up years is larger than or equal to n years and the patient did not die of lung cancer from death report. 2. The lung cancer risk is 1 if pathology confirmed lung cancer within n years and the follow-up years is large than or equal to n years. 3. The exam will be excluded otherwise.

Part of **Supplementary Table 6 Detail Comparison results in terms of AUC (%) and 95% CI**.

Task	Sybil*	M3FM (CT Only)	M3FM
1-Year CVD Mortality Risk	78.38 (57.09-100)	78.33 (56.4-100)	**84.05** (64.25-100)
2-Year CVD Mortality Risk	70.38 (55.52-85.25)	81.30 (68.08-94.52)	**83.72** (71.10-96.34)
3-Year CVD Mortality Risk	68.12 (57.97-78.27)	81.64 (72.77-90.51)	**85.89** (77.81-93.98)
4-Year CVD Mortality Risk	69.42 (60.87-77.98)	83.55 (76.31-90.8)	**88.30** (81.92-94.67)
5-Year CVD Mortality Risk	73.10 (66.00-80.19)	81.43 (75.03-87.83)	**86.13** (80.35-91.90)
6-Year CVD Mortality Risk	74.02 (67.47-80.57)	82.90 (77.10-88.69)	**87.71** (82.58-92.84)

Part of **Supplementary Table 7 Dataset distributions**. Numbers and percentages in parentheses of samples over candidate answers for all tasks in training, validation, and test datasets are reported.

Task	Answers	Train	Validation	Test
1-Year CVD Mortality Risk	0	73 (0.43%)	2 (0.08%)	4 (0.15%)
	1	16927 (99.57%)	2626 (99.92%)	2650 (99.85%)
	All	17000	2628	2654
2-Year CVD Mortality Risk	0	153 (0.90%)	6 (0.23%)	10 (0.38%)
	1	16847 (99.1%)	2622 (99.77%)	2644 (0.79%)
	All	17000	2628	2654
3-Year CVD Mortality Risk	0	241 (1.42%)	12 (0.46%)	21 (0.79%)
	1	16759 (98.58%)	2616 (99.54%)	2633 (99.21%)
	All	17000	2628	2654
4-Year CVD Mortality Risk	0	331 (1.95%)	27 (1.03%)	28 (1.06%)
	1	16669 (98.05%)	2601 (98.97%)	2626 (98.94)
	All	17000	2628	2654
5-Year CVD Mortality Risk	0	420 (2.47%)	34 (1.29%)	33 (1.24%)
	1	16580 (97.53%)	2594 (98.71%)	2621 (98.76%)
	All	17000	2628	2654
6-Year CVD Mortality Risk	0	478 (2.81%)	41 (1.56%)	36 (1.36%)
	1	16522 (97.19%)	2587 (98.44%)	2618 (98.64%)
	All	17000	2628	2654

2. The pre-trained imaging encoder can only take imaging patches of lungs and hearts as input. This should be considered as a limitation, as this will pose restrictions on other potential opportunistic assessment of body sections depicted in lung screening LDCT such as liver fat, bone mineral density, and body compositions (10.1148/radiol.222044, 10.1002/jbmr.4187, 10.1148/radiol.222937, 10.1016/j.clinimag.2019.12.009)

Response 1.2: Thanks for the important comment. By design, our imaging encoder is capable of processing multi-scale 3D CT images. In the self-supervised pretraining stage, our encoder was trained with augmented inputs that cover each whole CT volume on multi-scales. We have clarified in Sub-section 4.7 that "*In pretraining CTViT, the CT volume was randomly scaled by the factor of [0.5, 2], [0.6, 1.4], and [0.6, 1.4] in axial, coronal, and sagittal dimensions, with the voxel size accordingly calculated. Then, on each GPU a single input size was randomly chosen from a predefined set of input sizes. The input region of the chosen size was randomly cropped within the whole CT volume, not limited to the lung and heart regions.*" In practice, a radiologist only focuses on the lungs when reporting lung abnormalities, and on the heart when reporting heart problems. Since the labels in the NLST dataset are only for lungs and heart abnormalities, in the M3FM training stage our model paid main attention on the lungs and heart with the aligned labels, which saves computational cost by suppressing irrelevant regions, being consistent with radiologists' interpretation process. Nevertheless, the pretrained image encoder has no inherent limitation in accommodating opportunistic assessment due to the built-in multi-scale ability. We agree that the current version of M3FM was not optimized to cover all potential opportunistic findings because the current datasets have limited labels. We have clarified the limitations in the revised Discussions that "*Although the current M3FM models can perform a major set of LCS-related tasks covering all radiological labels in NLST, it is limited to predicting the abnormalities within the lungs and heart since the available labels are mainly targeted to these regions in our collected datasets. However, thanks to the built-in capability of multi-scale deep CT image analysis, it is straightforward to extend our M3FM models to handle other opportunistic findings*

[57–60] by focusing on the involved regions coupled with relevant clinical data and corresponding labels."

3. To encode the clinical meta data such as demographic, lifestyle, and disease diagnostic history, the authors created a clinical text following certain predefined rules and used this clinical text as condition to obtain the text embeddings forwarded to the task encoder. Since these clinical texts followed predefined rules, this fundamentally differed from the unstructured text in natural language and pose the question that whether or not this step is necessary comparing to simply encode these clinical metadata as an array.

Response 1.3: Thanks for this great question. We have now systematically studied different methods for encoding the clinical data. Specifically, we studied three encoding methods on the CVD mortality risk prediction tasks. The first method transforms clinical data into an array. The second method encodes clinical data as format-fixed text. The third method encodes clinical data as free-form text.

For the first method, we used the array of 83 elements to encode all the clinical data with five types: (1) category id (such as race, education, etc.), (2) age (such as patient age, disease age, etc.), (3) float numbers (weight, height, smoking years, etc.), (4) binary label (such as various disease and cancer histories), and (5) -1 (when the clinical data are not provided). Using the first method, the best results were achieved using a two-layer MLP with a dropout rate of 0.2 and then concatenating the embedding with the CT embedding. We also used the Text Transformer to encode the clinical array, which performed worse than the two-layer MLP.

For the second and third methods, we used the Text Transformer. To simulate free-form text, we used the LLAMA 3.1 405B large language model to convert each format-fixed clinical text to 20 different free-form versions of the same semantic meaning. Also, we studied the generalizability between using format-fixed or free-form text as training data and the other as testing data. The new results have been put in **Supplementary Table 9**.

Supplementary Table 9. Comparison of different methods for encoding clinical data. AUC (%) values with a 95% CI in parentheses for multi-year CVD mortality risk predictions are reported.

Train Input	LDCT	LDCT + Array	LDCT + Format-Fixed Text	CT + Format-Fixed Text	CT + Free-Form Text	CT + Free-Form Text
Test Input	LDCT	LDCT + Array	LDCT + Free-Form Text	CT + Free-Form Text	CT + Format-Fixed Text	CT + Free-Form Text
Mean	81.53	84.45	86.53	86.28	86.23	86.55
1-Year CVD Mortality Risk	78.33 (56.4-100)	82.70 (62.33-100)	83.59 (63.59-100)	84.14 (64.38-100)	89.28 (72.27-100)	87.84 (69.95-100)
2-Year CVD Mortality Risk	81.30 (68.08-94.52)	81.94 (68.87-95.02)	86.49 (74.71-98.28)	84.23 (71.74-96.71)	82.27 (69.28-95.27)	83.35 (70.63-96.07)
3-Year CVD Mortality Risk	81.64 (72.77-90.51)	85.10 (76.85-93.35)	86.07 (78.03-94.12)	86.90 (79.04-94.76)	85.59 (77.45-93.74)	87.51 (79.80-95.23)
4-Year CVD Mortality Risk	83.55 (76.31-90.8)	86.53 (79.80-93.26)	87.75 (81.26-94.24)	87.69 (81.19-94.19)	87.68 (81.18-94.18)	87.73 (81.23-94.22)
5-Year CVD Mortality Risk	81.43 (75.03-87.83)	84.33 (78.29-90.37)	86.93 (81.29-92.58)	86.56 (80.86-92.27)	85.54 (79.68-91.41)	85.85 (80.03-91.66)
6-Year CVD Mortality Risk	82.90 (77.10-88.69)	86.09 (80.71-91.47)	88.34 (83.32-93.36)	88.17 (83.12-93.22)	87.01 (81.78-92.25)	87.01 (81.77-92.25)

Encoding clinical data in both array and text improved the model performance relative to that using CT data only. The model achieved better results with the second and third methods (clinical text) than with the first method (array-shaped data). There is no significant performance difference when training with format-fixed text and testing with free-form text, or training with free-form text and testing with form-fixed text. Based on these new experimental results, our answer to your question is that encoding clinical data into either format-fixed or free-form text leads to better results than encoding clinical data into an array. The reason may be that the Text Transformer was pretrained on a large-scale text dataset and thus performed better when clinical data are presented as text tokens than an *ad hoc* network trained from scratch to encode the clinical data into an array.

All these results and analysis have been added into Supplementary Methods, and briefly described in Sub-section 4.2 (Text Transformer) "*Additionally, we comprehensively compared different methods for encoding clinical data in the format of an array, format-fixed text, and free-form text respectively. The results show that encoding clinical data with either format-fixed or free-form text achieved better results than that with array data. Also, encoding clinical data into format-fixed and free-form text achieved similar results. See Supplementary Methods and Supplementary Table 9 for details.*"

4. Fig.1 can be improved to clearly indicate the input / output for the training and inference/test steps. The dashed vertical line makes it confuse as to what types of data were used as training and inference.

Response 1.4: We appreciate this suggestion and have significantly revised **Figure 1** as follows.

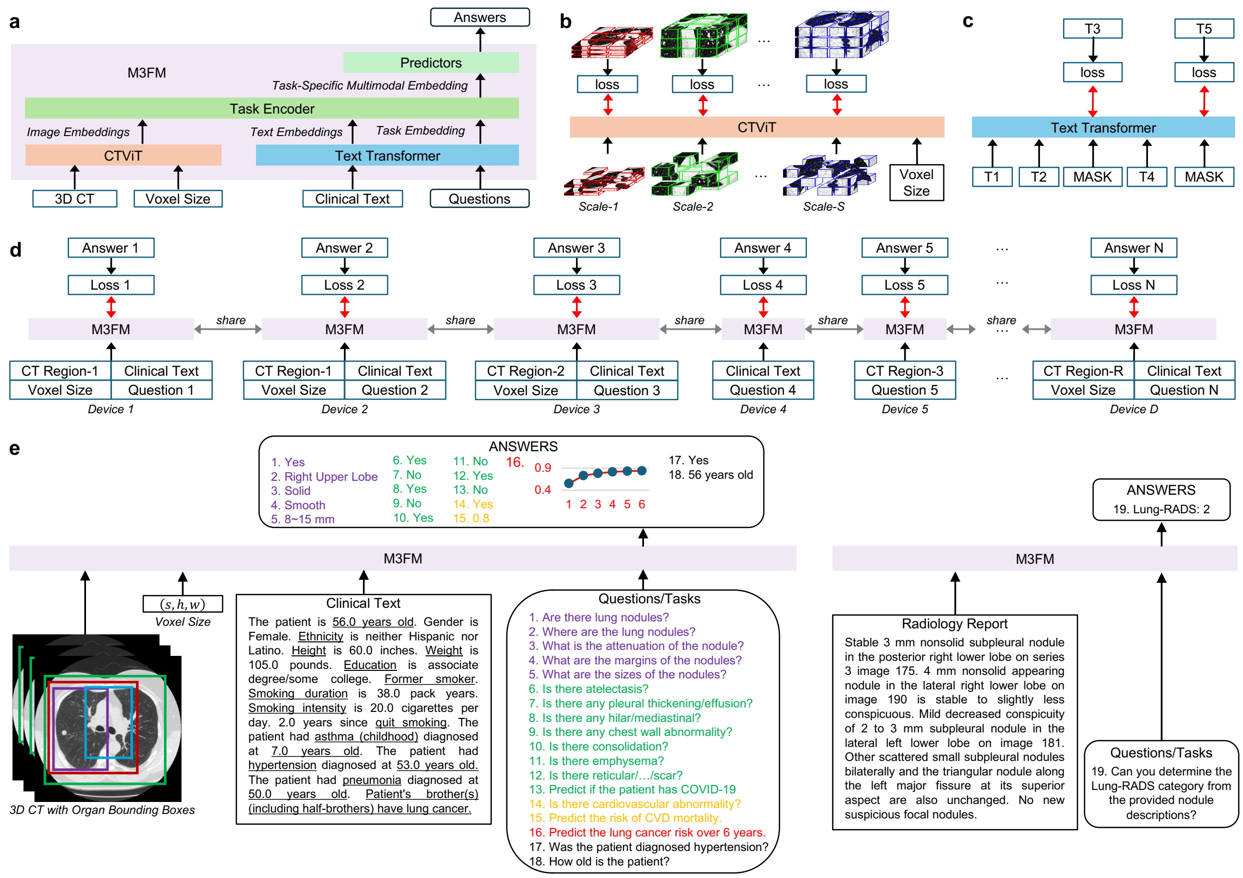


Figure 1. Overview of M3FM. a *M3FM architecture consists of four components: CTViT, Text Transformer, Task Encoder, and Predictors. CTViT encodes multi-scale 3D CTs. Text Transformer encodes clinical text and questions in free-form text. All image and text tokens are forwarded to Task Encoder, which extracts task-specific multimodal features. Predictors take task-specific features to derive the final answers. More details can be found in Figure 10.* ***b*** *Pretraining CTViT on extensive multiscale 3D CT volumes with voxel size-aware masked image modeling.* ***c*** *Pretraining Text Transformer with masked image models, where T1, T2, …, T5 denote a sequence of text tokens, T1, T2, and T4 are inputs, T3 and T5 are targets, and MASK is a special token meaning that the input token is masked.* ***d*** *Training the shared M3FM jointly with flexible multimodal and synergistic multitask learning using our distributed task-parallel (DTP) strategy. On each device, a single task is focused on with the task-specific inputs, targets, and loss functions. For example, different tasks may have the same multimodal inputs on devices 1 and 2 and various multi-/single-modality inputs on devices 2, 3, 4, and 5.* ***e*** *M3FM inference flexibly handles multi-scale CT volumes (indicated by the rectangle boxes in different sizes and colors), clinical data, and multiple tasks. The colors of the CT bounding boxes match those of the questions and the predicted answers. For example, to answer Question 16, M3FM takes the red region in the CT volume automatically localized using an organ localization model, the corresponding voxel size, and clinical text data as inputs. Questions 17 and 18 are two examples of auxiliary information*

retrieval tasks for clinical data modeling, which only take the clinical text as input. Question 19 predicts the Lung-RADS from lung nodule descriptions in a radiology report.

5. Line 384-385 – "We used the method developed in WFUSM as the reference": missing a reference.

Response 1.5: Thanks. We have cited the reference: *Lyu, Q., Yuan, H., Lin, Z., Ponnatapura, J., Whitlow, C.T.: A novel prediction model for immunotherapy induced pneumonitis prediction based on chest CT and electronic health record. medRxiv, 2024–10 (2024)*

6. Line 712-714: in addition to the number of V100 GPUs used, could you also provide the computing time?

Response 1.6: Thanks for the question. We have added the computing time in the revised manuscript: "*For CTViT pretraining and multi-task training, we used 192 NVIDIA Tesla V100 GPUs with 32 GiB of memory each (6 GPUs per node x 32 nodes). The CTViT pretraining took around 60 hours. The multi-task training took about 30 hours. For single-task training, we used 12 NVIDIA Tesla V100 GPUs with 32 GiB of memory each (6 GPUs per node x 2 nodes). The single-task training took about 22 hours*."

Point-by-Point Responses to Reviewer #2's Comments

Please note that all figure, table, and reference numbers mentioned correspond to those in the revised manuscript.

The manuscript describes the development of a lung cancer screening multimodal foundation model that learns, in a multitask setting, how to extract information from both CT scans and clinical reports in the form of text. First of all, I believe that the application of innovative deep learning breakthroughs in real-world and complex domains such as the clinical environment is extremely valuable, not only to assess how current models behave under real-world scenarios but also to more easily expose certain failure modes that are easily covered by typical large-scale training.

Response 2.0a: Thank you for your encouraging comment on the value of our work. A major purpose of this study is to develop and evaluate the most advanced large foundation models with the real-world multimodal multitask datasets from NSLT, MIDRC, and two quaternary hospitals. This scenario is complex and challenging as it involves 17 clinical tasks and 49 data types in a rather big data (~20TB). We have presented the whole workflow for the M3FM development, from LCS multitask definition to multimodal data curation, from radiologists' interpreting procedure to our unified MQA framework, and from self-supervised pre-training to multitask learning with high-dimensional multimodal data. In this revision, extensive experiments have been conducted to study the synergistic effects of multimodal data and multitask learning on each specific task, which is instrumental to understand the behaviors of large foundation models and pave the way towards clinical translation.

Considering the field's state-of-the-art, the authors' analysis focuses on the fact that most of the literature is limited to single-modality and single-task (SM-ST) approaches, which I agree. However, I think that a deeper elaboration on why including different data modalities and training for several tasks at once brings any benefits when compared with SM-ST counterparts is still missing (more direct concerns related to this in the points below). Overall, I found the methodology interesting, well-reported and with enough detail to ensure reproducibility.

Response 2.0b: Thank you so much for the important comment. We have done our best to follow your advice for deeper analysis on multimodal data and multitask learning. We have added more experimental results and given detailed discussions on "how" and "when" multimodal and multitask learning can be beneficial. The point-by-point responses are below.

1. Is it possible that, for certain questions, irrelevant input information is given for the model to provide a prediction? For instance, Table 1 shows that for questions like "Is there any lung nodule" or "Where is the lung nodule", several different sources of text inputs are given alongside the CT exam. In cases like this, isn't it enough to look at imaging information? The authors mention this subject in l.431, but it remains unclear whether you agree or not that different modalities should be provided even if they're not relevant a priori.

Response 2.1: Thanks for the insightful comment. The ablation studies in **Table 1** evaluated the effects of multimodal data on every task by comparing the performance metrics between training M3FM with LDCT scans only and training M3FM with multimodal datasets. To answer your questions effectively, we have further evaluated these models on multimodal inputs versus single-modality LDCT input, as summarized in the updated **Tables 2 and 3**, and new **Supplementary Table 7**. These tables are shown below for your convenience.

Specifically, we have compared the models that were trained and tested with different inputs for CVD diagnosis and CVD mortality risk prediction in **Table 2**, for lung cancer risk prediction in **Table 3**, and for other tasks in **Supplementary Table 8**. Based on these results, we have clarified in the Discussions part (here the direct answers to your questions are underlined): "*The results in Table 1 show that the multimodal data are particularly helpful for improving CVD diagnosis and*

CVD mortality risk prediction with M3FM. <u>However, some other tasks, such as lung nodule detection and characterization did not benefit from additional clinical data types, suggesting that the imaging data is enough for these tasks</u>. On the other hand, the evaluation results in Tables 2, 3 and Supplementary Table 8 show that training M3FM with additional clinical input data types would not degrade the performance in comparison with single-modality LDCT models on the tasks that were not benefited from the clinical priors. In other words, there are no constraining effects of multimodal learning on the representations of the single-mortality models trained with LDCT scans only. <u>Therefore, incorporating multimodal data as inputs is generally beneficial or at least does not harm model performance.</u> In this context, M3FM also provides a flexible framework to study the correlation between the model performance and different combinations of multimodal inputs on each specific task for optimized selection and modeling of multimodal datasets."

Table 2 Evaluation of multimodal elements in CVD tasks. AUC (%) values with a 95% CI in parentheses are reported.

Train Input	LDCT	LDCT + All Clinical Text									
Test Input	LDCT	LDCT	LDCT + All Clinical Text	LDCT + All Disease Text	LDCT + Heart Disease/Attack	LDCT + Hypertension	LDCT + Diabetes	LDCT + Stroke	LDCT + Heart Disease/Attack, Hypertension	LDCT + Heart Disease/Attack, Hypertension, Stroke	LDCT + Heart Disease/Attack, Stroke, Hypertension, Diabetes
CVD Diagnosis	89.24 (87.45-91.04)	88.46 (86.61-90.30)	92.38 (90.84-93.92)	92.37 (90.83-93.91)	90.01 (88.28-91.75)	91.31 (89.78-92.94)	89.29 (87.49-91.08)	88.97 (87.15-90.78)	92.21 (90.65-93.77)	92.27 (90.72-93.82)	92.46 (90.93-94.00)
CVD Mortality	81.63 (75.85-87.41)	83.44 (77.86-89.02)	87.09 (82.00-92.19)	87.09 (82.00-92.18)	83.27 (77.67-88.87)	86.41 (81.22-91.60)	84.73 (79.30-90.15)	84.54 (79.10-89.99)	86.89 (81.77-92.01)	86.84 (81.71-91.97)	87.29 (82.23-92.35)

Table 3 Evaluation of multimodal elements in lung cancer risk prediction. AUC (%) values with a 95% CI in parentheses are reported.

Train input	LDCT	LDCT+ All Clinical Text					
Test input	LDCT	LDCT	LDCT+ All Clinical Text	LDCT + Demographic	LDCT + Smoking History	LDCT + Cancer History	LDCT + Disease History
Mean	84.60	85.43	85.22	85.62	85.42	85.35	85.01
1-Year Lung Cancer Risk	92.98 (89.80-96.15)	93.54 (90.47-96.60)	93.62 (90.58-96.66)	93.51 (90.44-96.58)	93.54 (90.47-96.60)	93.68 (90.65-96.71)	93.38 (90.28-96.47)
2-Year Lung Cancer Risk	86.97 (83.58-90.36)	87.17 (83.80-90.54)	87.27 (83.91-90.63)	87.23 (83.87-90.60)	87.17 (83.80-86.63)	87.28 (83.92-90.63)	87.00 (83.61-90.39)
3-Year Lung Cancer Risk	84.18 (80.88-87.48)	85.10 (81.87-88.33)	84.79 (81.54-88.05)	85.21 (81.99-88.44)	85.09 (81.86-88.32)	85.01 (81.77-88.24)	84.56 (81.29-87.83)
4-Year Lung Cancer Risk	83.38 (80.20-86.55)	83.47 (80.30-86.64)	82.99 (79.79-86.19)	83.69 (80.54-86.84)	83.46 (80.29-86.63)	83.20 (80.01-86.39)	82.90 (79.69-86.11)
5-Year Lung Cancer Risk	80.88 (77.73-84.02)	81.75 (78.65-84.85)	81.31 (78.19-84.43)	82.10 (79.02-85.17)	81.73 (78.63-84.83)	81.55 (78.44-84.66)	81.08 (77.94-84.21)
6-Year Lung Cancer Risk	79.22 (76.06-82.38)	81.57 (78.53-84.61)	81.35 (78.29-84.40)	81.95 (78.93-84.98)	81.56 (78.51-84.60)	81.40 (78.35-84.45)	81.13 (78.07-84.20)

Supplementary Table 8 Ablation study on multimodal inputs in some other tasks*. In terms of AUC (\%) and 95\% CI results, the multimodal inputs achieve equivalent or slightly better performance as compared to that with the single-modality LDCT input (the second, third and fourth columns). Generally, multimodal learning does not degrade the model performance obtained through LDCT single-modality learning (the second and third columns).*

Train Input	LDCT	LDCT + Clinical Text	LDCT + Clinical Text
Test Input	LDCT	LDCT	LDCT + Clinical Text
Nodule Presence	98.77 (98.63-98.92)	98.73 (98.58, 98.88)	98.76 (98.62-98.91)
Nodule Location (Average AUC)	98.84	98.93	98.93
Right Upper Lobe (RUL)	99.12 (98.87-99.37)	99.15 (98.91-99.39)	99.15 (98.91-99.39)
Right Middle Lobe (RML)	98.11 (97.60-98.62)	98.26 (97.77-98.75)	98.26 (97.77-98.75)
Right Lower Lobe (RLL)	98.56 (98.24-98.87)	98.65 (98.35-98.95)	98.65 (98.35-98.95)
Left Upper Lobe (LUL)	99.18 (98.92-99.44)	99.28 (99.04-99.52)	99.28 (99.04-99.52)
Left Lower Lobe (LLL)	99.22 (98.97-99.46)	99.32 (99.09-99.55)	99.32 (99.09-99.55)
Nodule attenuation (Average AUC)	75.40	75.24	75.25
Solid	78.17 (77.28-79.06)	77.99 (77.11-78.88)	78.00 (77.11-78.89)
Ground Glass	84.10 (82.85-85.34)	83.36 (82.09-84.61)	83.37 (82.10-84.63)
Others	63.93 (61.95-65.92)	64.37 (62.38-66.37)	64.37 (62.38-66.36)
Nodule margin (Average AUC)	76.37	76.48	76.53
Spiculated	79.29 (77.60- 80.97)	79.01 (77.32-80.71)	78.93 (77.24-80.62)

Smooth	78.92 (78.11-79.74)	77.81 (76.98-78.65)	78.05 (77.22-78.88)
Poorly Defined	77.50 (76.28-78.71)	74.91 (73.66-76.17)	75.11 (73.86-76.36)
Undetermined	69.75 (66.67-72.84)	74.17 (71.18-77.16)	74.04 (71.05-77.03)
Nodule size	82.30	81.51	81.67
<=4 mm	77.32 (76.24-78.39)	77.79 (76.72-78.86)	77.60 (76.53-78.67)
4~6 mm	70.06 (69.08-71.04)	67.98 (66.98-68.98)	67.82 (66.82-68.82)
6~8 mm	75.42 (74.10-76.74)	73.36 (72.01-74.71)	73.30 (71.95-74.65)
8~15 mm	86.81 (85.56-88.07)	85.84 (84.55-87.13)	85.95 (84.66-87.23)
15~30 mm	91.02 (89.21-92.84)	90.71 (88.87-92.55)	91.25 (89.46-93.05)
>30 mm	93.16 (89.14-97.18)	93.40 (89.44-97.36)	94.13 (90.38-97.89)
Atelectasis	81.72 (77.23-86.22)	81.09 (76.55-85.64)	81.08 (76.54-85.63)
Pleural thickening/effusion	73.73 (71.78-75.67)	75.35 (73.43-77.26)	76.07 (74.17-77.97)
Hilar/mediastinal adenopathy/mass	82.99 (79.85-86.13)	82.94 (79.80-86.08)	82.97 (79.83-86.11)
Chest wall abnormality	81.51, (66.14-96.89)	83.36 (68.24-98.48)	82.39 (68.59-97.18)
Consolidation	65.15 (59.39-70.92)	68.43 (62.74-74.12)	68.95 (63.26-74.64)
Emphysema	91.37 (90.79-91.94)	92.39 (91.85-92.93)	92.40 (91.86-92.94)
Reticular/reticulonodular opacities/honeycombing/fibrosis/scar	76.76 (75.73-77.79)	78.88 (77.89-79.88)	79.29 (78.30-80.27)
COVID-19	76.88 (75.53-78.23)	76.55 (75.19-77.90)	76.79 (75.44-78.14)

2. Are there any implications that result from treating clinical information that possibly comes from tabular data (hence being explicitly symbolic and represented in a non-ambiguous way) as free text? For instance, can your specific sentence format result in some sort of overfitting in the way that different sentences with equivalent meaning would cause the model to fail?

Response 2.2: Thanks for this great question. We have now systematically studied different methods for encoding the clinical data. Specifically, we studied three encoding methods on the CVD mortality risk prediction tasks. The first method transforms clinical data into an array. The second method encodes clinical data as format-fixed text. The third method encodes clinical data as free-form text.

For the first method, we used the array of 83 elements to encode all the clinical data with five types: (1) category id (such as race, education, etc.), (2) age (such as patient age, disease age, etc.), (3) float numbers (weight, height, smoking years, etc.), (4) binary label (such as various disease and cancer histories), and (5) -1 (when the clinical data are not provided). Using the first method, the best results were achieved using a two-layer MLP with a dropout rate of 0.2 and then concatenating the embedding with the CT embedding. We also used the Text Transformer to encode the clinical array, which performed worse than the two-layer MLP.

For the second and third methods, we used the Text Transformer. To simulate free-form text, we used the LLAMA 3.1 405B large language model to convert each format-fixed clinical text to 20 different free-form versions of the same semantic meaning. Here is an exemplary format-fixed text "*The patient is 67 years old. Gender is Male. Race is white. Ethnicity is neither hispanic nor latino. Height is 72 inches. Weight is 182 pounds. Education is associate degree/ some college. Current smoker. Smoking duration is 40 pack years. Smoking intensity is 20 cigarettes per day. The patient had asthma (adult) diagnosed at 58 years old. The patient had hypertension diagnosed at 68 years old.*" It was converted to various free-form text versions with the same semantic meaning; for example, "*This patient is male, 67 years old, white with non-Hispanic and non-Latino background, holds an associate degree, and smokes 20 cigarettes daily (40 pack years). Diagnosed with asthma at 58 and hypertension at 68, he is 72 inches tall and weighs 182 pounds.*" The new results have been put in Supplementary **Table 9**.

Supplementary Table 9. Comparison of different methods for encoding clinical data. AUC (%) values with a 95% CI in parentheses for multi-year CVD mortality risk predictions are reported.

Train Input	LDCT	LDCT + Array	LDCT + Format-Fixed Text	CT + Format-Fixed Text	CT + Free-Form Text	CT + Free-Form Text
Test Input	LDCT	LDCT + Array	LDCT + Free-Form Text	CT + Free-Form Text	CT + Format-Fixed Text	CT + Free-Form Text
Mean	81.53	84.45	86.53	86.28	86.23	86.55
1-Year CVD Mortality Risk	78.33 (56.4-100)	82.70 (62.33-100)	83.59 (63.59-100)	84.14 (64.38-100)	89.28 (72.27-100)	87.84 (69.95-100)
2-Year CVD Mortality Risk	81.30 (68.08-94.52)	81.94 (68.87-95.02)	86.49 (74.71-98.28)	84.23 (71.74-96.71)	82.27 (69.28-95.27)	83.35 (70.63-96.07)
3-Year CVD Mortality Risk	81.64 (72.77-90.51)	85.10 (76.85-93.35)	86.07 (78.03-94.12)	86.90 (79.04-94.76)	85.59 (77.45-93.74)	87.51 (79.80-95.23)
4-Year CVD Mortality Risk	83.55 (76.31-90.8)	86.53 (79.80-93.26)	87.75 (81.26-94.24)	87.69 (81.19-94.19)	87.68 (81.18-94.18)	87.73 (81.23-94.22)
5-Year CVD Mortality Risk	81.43 (75.03-87.83)	84.33 (78.29-90.37)	86.93 (81.29-92.58)	86.56 (80.86-92.27)	85.54 (79.68-91.41)	85.85 (80.03-91.66)
6-Year CVD Mortality Risk	82.90 (77.10-88.69)	86.09 (80.71-91.47)	88.34 (83.32-93.36)	88.17 (83.12-93.22)	87.01 (81.78-92.25)	87.01 (81.77-92.25)

Encoding clinical data in both array and text improved the model performance relative to that using CT data only. The model achieved better results with the second and third methods (clinical text) than with the first method (array-shaped data). There is no significant performance difference when training with format-fixed text and testing with free-form text, or training with free-form text and testing with form-fixed text. Based on these new experimental results, our answer to your question is that encoding clinical data into either format-fixed or free-form text leads to better results than encoding clinical data into an array. The reason may be that the Text Transformer was pretrained on a large-scale text dataset and thus performed better when clinical data are presented as text tokens than an *ad hoc* network trained from scratch to encode the clinical data into an array.

Also, we studied the generalizability between using format-fixed or free-form text as training data and the other as testing data. Interestingly, we found that even when the model was trained exclusively on format-fixed text, its testing performance with free-form text was comparable to that with format-fixed text as input, and vice versa. This aligns with prior empirical findings, suggesting that pretraining a Text Transformer on a large-scale text dataset can help prevent overfitting and enhance performance on downstream tasks through fine-tuning.

All these results and analysis have been added into Supplementary Methods, and briefly described in Sub-section 4.2 (Text Transformer) "*Additionally, we comprehensively compared different methods for encoding clinical data in the format of an array, format-fixed text, and free-form text respectively. The results show that encoding clinical data with either format-fixed or free-form text achieved better results than that with array data. Also, encoding clinical data into format-fixed and free-form text achieved similar results. See Supplementary Methods and Supplementary Table 9 for details.*"

3. The reported results demonstrate that, overall, AUC values increase for MT-MM variants. However, I think the authors could give a more critical perspective when balancing the advantages in terms of performance values - which in some cases are small - with the extra effort of processing multiple datasets and creating adequate tasks.

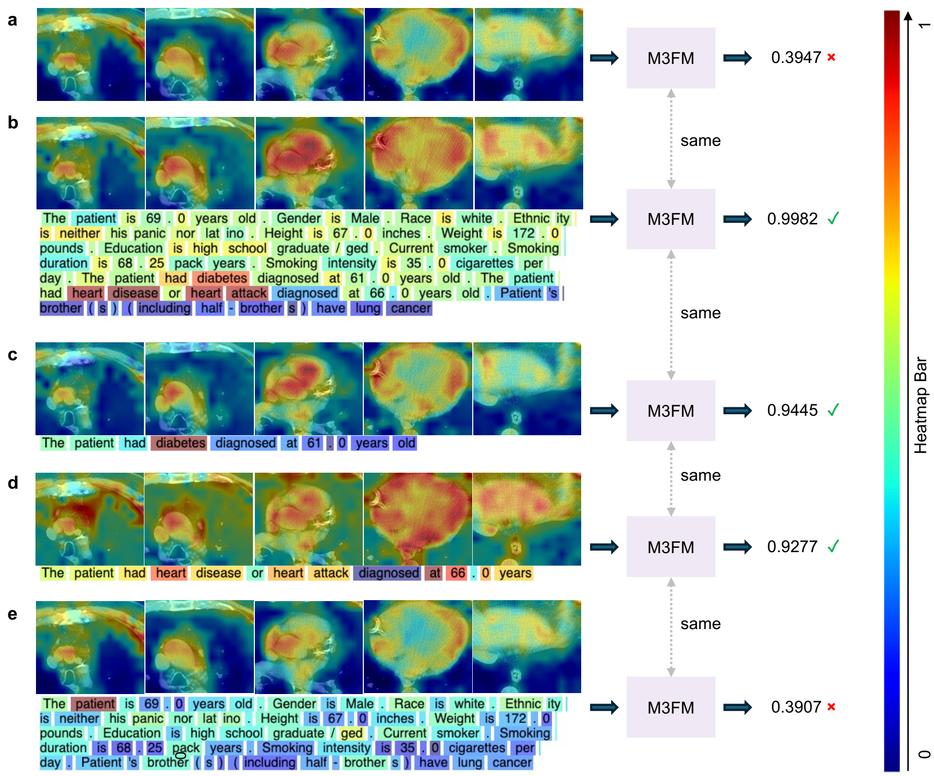
Response 2.3: Thanks for the insightful comment. Based on the comparative results in **Table 1**, we have given an analysis in the Discussions part that "Overall, the results in Table 1 demonstrate that M3FM optimized with multitask learning outperformed the corresponding models that were separately optimized for individual tasks. These results indicate that for intrinsically-related tasks if their multimodal data and clinical labels are simultaneously collected, such as in the NSLT trail for LCS, M3FM can synergistically integrate multiple tasks that may take various scales of imaging data and different multimodal inputs for improved performance using our proposed distributed task-parallel training approach. However, when non-trivial efforts are required to collect additional multitask datasets for a particular task at hand, there would be a trade-off between performance improvement (especially if the improvement is marginal) and the associated cost of constructing additional datasets. In the latter case, M3FM provides a useful framework to study this trade-off by performing ablation studies as shown in Table 1 on relatively small pilot datasets prior to collecting large-scale datasets."

4. The heatmaps in Figure 6 show that the presence of different text tokens is "important" for the model's decision, even though some of them are apparently useless by themselves (e.g. "brother", "142"). I think the authors should analyze this issue when describing the figure.

Response 2.4: Thanks for the suggestion. Here we would like to clarify that the family lung cancer history was defined as a risk factor in NLST, such as "brother(s) (including half-brother(s)) has lung cancer". We agree that some less relevant tokens were highlighted for the qualitative examples in Figure 6. Following the reviewer's suggestion, we have underlined the issue in Sub-section 2.4 that "*Although the visualization of attention maps provides a window to inspect the behavior of the Transformer model, it is not always reliable to reveal the correlation between model predictions and input tokens, e.g., less relevant tokens were highlighted in Figure 7, which is consistent to prior findings [52, 53]*."

5. Even though the analogy to human learning processes can be easily raised, can it be explicitly shown that information from multiple sources is actually being composed and learned in a complementary way? Is there any constraining effect on the kind of representations that are captured from image-based models when encoding multiple sources of information in parallel instead of only using images? When performing multiple tasks, is it possible to demonstrate that pieces of knowledge are being reused? If so, which ones? I think questions like these should be deeply explored in approaches like M3FM.

Response 2.5: We appreciate your insightful comments. To visualize how the information from multiple sources is learned and composed, in addition to our statistically established results from the ablation studies, we have added a new **Supplementary Fig. 1** shown below. The corresponding explanation has been added in Sub-section 2.4 that "*Furthermore, the ablation inference in Supplementary Fig. 1 explicitly shows how the information from multiple sources is composed to affect the model prediction. In a case of positive CVD diagnosis, M3FM had failed predictions when taking LDCT only or LDCT plus uninformative clinical data as inputs. The same M3FM successfully diagnosed CVD when using LDCT plus the relevant clinical data including the diabetes/heart disease history. This is consistent with the ablation results summarized in Table 2, where multimodal inputs are statistically beneficial for the M3FM inferences*."



Supplementary Fig. 1: **Ablation study on M3FM inference**. Different input combinations of a clinically positive example were separately forwarded to the same M3FM for the CVD diagnosis. Attention maps are overlapped on the multimodal inputs. **a**. M3FM predicted a low score with the single-mortality LDCT input. **b**. M3FM predicted the highest score using the multimodal inputs including LDCT and all clinical data. **c**. M3FM predicted a high score with the LDCT data and diabetes history. **d**. M3FM predicted a high score with the LDCT data and heart disease history. **e**. M3FM predicted a low score with the LDCT data and all clinical data excluding the diabetes/heart disease history.

To study if there is constraining effect on the representations captured from the image-based models when encoding multiple sources of information simultaneously, we have compared the performance metrics of these models with LDCT data as input only versus multimodal data as inputs in the training stage, and then using LDCT data only as the input in the testing stage for all tasks. The corresponding results are reported in the updated **Tables 2 and 3**, and new **Supplementary Table 8**, as shown under **Response 2.1**. We have clarified in the Discussions part that "…*On the other hand, the results in Tables 2, 3 and Supplementary Table 8 show that training M3FM with additional clinical input data would not degrade the performance in comparison with the single-modality LDCT models on the tasks that were not benefited from the clinical priors. In other words, there are no constraining effects of multimodal learning on the representations of the single-mortality models trained with LDCT scans only...*"

Multitask learning is to jointly learn multiple related tasks for improved performance. Some theoretical studies [55, 56] have demonstrated how multitask learning is fundamentally advantageous over single-task learning through analysis on upper error bounds conditioned on the number of multiple tasks and the average number of data points per task. However, it is still an active area and challenging to explicitly show which pieces of knowledge are reused by a deep neural network among different tasks. Importantly, our empirical results in the ablation studies have shown that multitask learning generally achieved better generalizability on both testing and independent evaluation datasets in the real-world LCS applications, which are consistent with the aforementioned theoretical findings.

During our further analysis of our extensive results in the revision stage, we have made a very interesting observation that multitask learning is more beneficial for imbalanced datasets. Specifically, these results have been described in the Results section (Sub-section 2.3) that "*Impressively, training on multiple tasks, M3FM-MT-MM outperforms the M3FM-ST-MM for seventeen out of 22 (sub)-tasks. In reference to the label distributions of the multiple tasks in Supplementary Table 7, the five tasks that were not benefited from multitask learning have the largest balance ratios of the number of minority class samples over the number of majority class samples. In other words, multitask learning is more beneficial for tasks with more imbalanced datasets or a much smaller number of positive/minority class labels*."

Specific comments have been added in the Discussions: "*Theoretical studies [55,56] have indicated why learning multiple tasks jointly is beneficial over learning each task in isolation through analysis of the upper error bound conditioned on the number of multiple tasks and the average number of data points per task. In the real-world LCS applications, our results have shown that multitask learning generally achieved better generalizability on both the NLST test datasets and independently collected evaluation datasets, which are consistent with the theoretical findings. Interestingly, the empirical results further suggest that the LCS tasks benefited from multitask learning have more imbalanced labels. This seems heuristic, since when labels are sparse for a specific task, more synergy should be leveraged by learning from other related tasks. In other words, multitask learning promises to alleviate the overfitting and improve the generalizability via a regularization effect especially when the task has a limited number of samples [55,56]. Given that many clinical tasks involve highly imbalanced datasets with few labels of positive or certain type disease, our results mean that multitask learning is favorable in building foundation models in real-world scenarios*."

6. Finally, the ablation studies performed are valuable, and the comparison with different models also highlights situations where performance gaps are larger and smaller, which is useful.

Response 2.6: Thank you so much!

In general, even though the authors conducted several experiments, I think that showing a deeper analysis of the behavior of an MT-MM approach besides quantitative performance values is necessary to conclude that these training settings bring the benefits that are raised when motivating.

Response 2.7: Thanks for the constructive comments. Following your advice, we have conducted more experiments and performed deeper analysis on multimodal and multitask learning as described above. We hope these findings would motivate the community to build more advanced large foundation models for LCS and in other clinical scenarios.

Point-by-Point Responses to Reviewer #3's Comments

This manuscript details the development of multimodal multitask foundation model in the setting of lung cancer screening with low dose chest CT. The model is able to perceive multimodal data from CT and other clinical data sources and perform various tasks within the LCS workflow--including lung nodule detection and characterization, lung cancer risk prediction, cardiovascular disease diagnosis and mortality prediction, and Lung-RADS reporting/categorization, among others mimicking the multiple tasks a radiologist reading a LDCT for LCS screening would typically perform.

Response 3.1: We deeply appreciate your insightful summary and encouraging feedback on our work. We are excited about the remarkable capabilities of our multimodal multitask foundation model in the context of lung cancer screening. Our current goal is to develop and translate AI foundation models in the multiple tasks a radiologist performs during LDCT interpretation.

The model was developed from robust data sources (NLST and MIDRC) and tested on data from two large medical centers. The authors report that the model outperforms previous state-of-the-art models, including the *Sybil* models.

Response 3.2: We feel privileged to have the opportunity to redefine the state of the art in this critical clinical area. Multimodal big data, advanced network architecture, parallel supercomputing power, as well as rigorous evaluation and validation are indispensable and synergistic in this large-scale multidisciplinary project. In this direction, we are committed to advancing this work persistently with the hope that low-dose CT lung cancer screening tests can become more improved and more accessible, particularly in underdeveloped regions where radiologists are scarce.

The manuscript is overall well-written. Methodology is sound and results and discussion are robust and appropriate. The model has promise in leveraging AI in optimizing lung cancer screening implementation and other multimodal/multitask programs in the future.

Response 3.3: Thanks for your recognition of the manuscript's clarity and the soundness of our methodology. We share your vision that foundation models hold great potential for other multimodal multitask programs.